

Erdős Problem #1110

EP-1110

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $p > q \geq 2$ be two coprime integers. We call n representable if it is the sum of integers of the form $p^k q^l$, none of which divide each other.

If $\{p, q\} \in \{2, 3\}$ then what can be said about the density of non-representable numbers? Are there infinitely many coprime non-representable numbers?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>